POWER PULSE

Load Flow Studies Important Formulae

Notes by Balaji Sir

1 Bus types and Y-bus

1. $$\text{Slack: } |V|, \delta \text{ specified;} \quad \text{PV: } P, |V|; \quad \text{PQ: } P, Q$$

2. $$Y_{ii} = \sum_j y_{ij} + y_{i0}, \qquad Y_{ij} = -y_{ij}$$

off-diagonal = negative of the line admittance

3. $$P_i - jQ_i = V_i^* \sum_k Y_{ik} V_k$$

2 Gauss–Seidel

4. $$V_i^{(m+1)} = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{V_i^{(m)*}} - \sum_{k \neq i} Y_{ik} V_k \right]$$

5. $$V_{acc} = V_{old} + \alpha \left(V_{new} - V_{old} \right), \quad \alpha \approx 1.4\text{–}1.7$$

acceleration factor

6. $$\text{PV bus: keep the angle, reset } |V| \text{ to the schedule}$$

3 Newton–Raphson and decoupled

7. $$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ J_3 & J_4 \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta |V| \end{bmatrix}$$

Jacobian size = (n−1) + n_PQ

8. $$\text{Equations: } 2(n-1) \text{ for PQ buses only}$$

9. $$[B'] \Delta \delta = \frac{\Delta P}{|V|} \, (n-1), \qquad [B''] \Delta |V| = \frac{\Delta Q}{|V|} \, (n_{PQ})$$

fast decoupled: R ≪ X, small δ, |V| ≈ 1

4 DC load flow and line flows

10. $$P_{ik} = \frac{\delta_i - \delta_k}{X_{ik}}, \qquad [P] = [B'][\delta]$$

11. $$S_{ik} = V_i I_{ik}^*, \qquad S_{loss} = S_{ik} + S_{ki}$$

the slack bus supplies the system losses

12. $$\text{PV bus beyond } Q_{max} \Rightarrow \text{ treat as PQ with } Q = Q_{max}$$